Time series: Fitting the ARMA model

Andrey Pepelyshev

MA3508, MA4508, MAT508
Lecture notes 11

MATHS, Cardiff
Spring semester, 2026

The homework is provided on the last slide

The ARMA process with mean μ

Recall that x_t is an ARMA process with mean μ if $x_t - \mu$ is an ARMA process with mean zero, see slide 2 of Lecture 8.

Task. Compute the mean of an ARMA process given by $x_t = 0.2 + 0.4x_{t-1} + 0.5x_{t-2} + \epsilon_t + 0.7\epsilon_{t-1}$.

Taking the expectation of the given equation, we have

$$\mathbf{E}x_t = 0.2 + 0.4\mathbf{E}x_{t-1} + 0.5\mathbf{E}x_{t-2} + \mathbf{E}\epsilon_t + 0.7\mathbf{E}\epsilon_{t-1}.$$

Since $\mathbf{E}x_j = \mu$ and $\mathbf{E}\epsilon_j = 0$ for all j, we have

$$\mu = 0.2 + 0.4\mu + 0.5\mu.$$

Therefore,

$$\mu = \frac{0.2}{1 - 0.4 - 0.5} = \frac{0.2}{0.1} = 2.$$

Estimation of parameters of AR process

Let $x_1, \ldots, x_n$ be a realization of AR(p) process with mean μ.

We want to estimate the mean μ and coefficients $a_1, \ldots, a_p$ of a process $x_t - \mu = a_1(x_{t-1} - \mu) + \ldots + a_p(x_{t-p} - \mu) + \epsilon_t$.

First, we estimate μ by the average $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} x_j$.
Second, we estimate the correlation function

$$\hat{\rho}(k) = \frac{\hat{R}(k)}{\hat{R}(0)} = \frac{1/n \sum_{j=1}^{n-k}(x_j - \bar{X}_n)(x_{j+k} - \bar{X}_n)}{1/n \sum_{j=1}^{n}(x_j - \bar{X}_n)^2}.$$

Third, we estimate coefficients $a_1, \ldots, a_p$ from the system of Yule-Walker equations

$$\begin{cases} \hat{\rho}(1) = a_1\hat{\rho}(0) + a_2\hat{\rho}(1) + \ldots + a_p\hat{\rho}(1-p) \\ \hat{\rho}(2) = a_1\hat{\rho}(1) + a_2\hat{\rho}(0) + \ldots + a_p\hat{\rho}(2-p) \\ \vdots \\ \hat{\rho}(p) = a_1\hat{\rho}(p-1) + a_2\hat{\rho}(p-2) + \ldots + a_p\hat{\rho}(p-p) \end{cases}$$

Matrix notation of estimator

The estimator $\hat{a} = (\hat{a}_1, \ldots, \hat{a}_p)^T$ of coefficients of an AR(p) process is a solution of the system of Yule-Walker equations,

$$\hat{a} = \hat{C}^{-1}\hat{v},$$

where $\hat{v} = (\hat{\rho}(1), \hat{\rho}(2), \ldots, \hat{\rho}(p))^T$

$$\hat{C} = \Big(\hat{\rho}(i-j)\Big)_{i,j=1}^{p} = \begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \hat{\rho}(2) & \ldots & \hat{\rho}(p-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \hat{\rho}(1) & \ldots & \hat{\rho}(p-2) \\ \hat{\rho}(2) & \hat{\rho}(1) & \hat{\rho}(0) & \ldots & \hat{\rho}(p-3) \\ \vdots & \vdots & \vdots & & \vdots \\ \hat{\rho}(p-1) & \hat{\rho}(p-2) & \hat{\rho}(p-3) & \ldots & \hat{\rho}(0) \end{pmatrix}.$$

NB. *Compare with slide 6 of Lecture 6 and beware of the distinction.*

Estimating the variance of noise

Theorem. The estimator of the variance of x_t is given by $\hat{R}(0)$ and the estimator of the variance of ϵ_t is

$$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \ldots - \hat{a}_p\hat{\rho}(p))$$

Substituting the AR(p) equation into the expression, we get

$$\begin{aligned} R(0) &= \mathbf{E}(x_t - \mu)(x_t - \mu) \\ &= \mathbf{E}(x_t - \mu)(a_1(x_{t-1} - \mu) + \ldots + a_p(x_{t-p} - \mu) + \epsilon_t) \\ &= a_1R(1) + \ldots + a_pR(p) + \sigma^2 \\ &= R(0)[a_1\rho(1) + \ldots + a_p\rho(p)] + \sigma^2. \end{aligned}$$

Thus, $\sigma^2 = R(0)[1 - a_1\rho(1) - \ldots - a_p\rho(p)]$.

R code for estimation of AR(p) process

Task. Let $1, 4, 2, 7, 3, 1, 5, 2, 6$ be a realization of AR(3) process. Estimate parameters of this process.

```
® x=c(1,4,2,7,3,1,5,2,6); p=3;
n=length(x); mu=sum(x)/n; y=x-mu;
covg=rep(0,n); corrg=rep(0,n);
for(i in (1:n))
{
  covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n;
  corrg[i]=covg[i]/covg[1];
}
C=matrix(0,p,p); v=rep(0,p);
for(i in (1:p)) v[i]=corrg[i+1];
for(i in (1:p)) { for(j in (1:p)) C[i,j]=corrg[abs(i-j)+1];}
a=solve(C) %*% v; print(a);
sigmasq=covg[1]*(1-v %*% a);
```

Estimation of AR(1) process

Consider an AR(1) process with mean μ

$$x_t - \mu = a_1(x_{t-1} - \mu) + \epsilon_t$$

where ϵ_t is white noise with zero mean and variance σ^2.
For $p = 1$, the matrix $\hat{C}$ is of size 1×1, $\hat{C} = \hat{\rho}(0) = 1$, and the vector $\hat{v}$ of length 1, $\hat{v} = \hat{\rho}(1)$.

Theorem. The estimators for the AR(1) process have the form

$$\hat{a}_1 = \hat{\rho}(1) \quad \text{and} \quad \hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1)).$$

Moreover, $n\mathrm{Var}(\hat{a}_1) \to 1 - a^2$ as the length of a time series $n \to \infty$.

Estimation of AR(2) process

Consider an AR(2) process with mean μ

$$x_t - \mu = a_1(x_{t-1} - \mu) + a_2(x_{t-2} - \mu) + \epsilon_t$$

where ϵ_t is white noise with zero mean and variance σ^2.
For $p = 2$, we have

$$\hat{C} = \begin{pmatrix} 1 & \hat{\rho}(1) \\ \hat{\rho}(1) & 1 \end{pmatrix}, \quad \hat{v} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \end{pmatrix}.$$

Theorem. The estimators for the AR(2) process have the form

$$\hat{a}_1 = \hat{\rho}(1)\frac{1 - \hat{\rho}(2)}{1 - \hat{\rho}^2(1)}, \quad \hat{a}_2 = \frac{\hat{\rho}(2) - \hat{\rho}^2(1)}{1 - \hat{\rho}^2(1)}.$$

and

$$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)).$$

Example AR(1)

Task. Let $2, 6, 3, 7, 2$ be a time series from an AR(1) process. Estimate μ, a_1 and σ^2.

The estimator of the mean is $\hat{\mu} = \bar{X}_5 = (2+6+3+7+2)/5 = 4$. Then

t	$x_t - \bar{X}_n$	$x_{t+1} - \bar{X}_n$
1	-2	2
2	2	-1
3	-1	3
4	3	-2
5	-2	

$$\hat{R}(0) = [(-2)^2 + 2^2 + (-1)^2 + 3^2 + (-2)^2]/5 = 22/5$$
$$\hat{R}(1) = [(-2)2 + 2(-1) + (-1)3 + 3(-2)]/5 = -15/5, \ \hat{\rho}(1) = -15/22.$$

Then $\hat{a}_1 = \hat{\rho}(1) = -15/22$ and

$$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1)) = \frac{22}{5}(1 - (-\frac{15}{22})(-\frac{15}{22})) =$$
$$= \frac{22}{5} \cdot \frac{22^2 - 15^2}{22^2} = \frac{22^2 - 15^2}{5 \cdot 22} = 2.35.$$

Example AR(2)

Task. Let $2, 4, 1, 6, 2$ be a time series from an AR(2) process. Estimate μ, a_1, a_2 and σ^2.

The estimator of the mean is $\hat{\mu} = \bar{X}_5 = (2+4+1+6+2)/5 = 3$. Then

t	$x_t - \bar{X}_n$	$x_{t+1} - \bar{X}_n$	$x_{t+2} - \bar{X}_n$
1	-1	1	-2
2	1	-2	3
3	-2	3	-1
4	3	-1	
5	-1		

$\hat{R}(0) = [(-1)^2 + 1^2 + (-2)^2 + 3^2 + (-1)^2]/5 = 16/5$

$\hat{R}(1) = [(-1)1 + 1(-2) + (-2)3 + 3(-1)]/5 = -12/5, \ \hat{\rho}(1) = -12/16,$

$\hat{R}(2) = [(-1)(-2) + 1 \cdot 3 + (-2)(-1)]/5 = 7/5, \ \hat{\rho}(2) = 7/16,$ Then

$\hat{a}_1 = \hat{\rho}(1)\frac{1-\hat{\rho}(2)}{1-\hat{\rho}^2(1)} = -12/16\frac{1-7/16}{1-12^2/16^2} = -\frac{12(16-7)}{16^2-12^2} = -\frac{12\cdot 9}{16^2-12^2} = -0.96$

$\hat{a}_2 = \frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)} = \frac{7/16-12^2/16^2}{1-12^2/16^2} = \frac{16\cdot 7-12^2}{16^2-12^2} = -0.29$

$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)) = \frac{16}{5}(1 - \frac{-12\cdot 9}{16^2-12^2}\frac{-12}{16} - \frac{16\cdot 7-12^2}{16^2-12^2}\frac{7}{16}) =$

$= \frac{1}{5}(16 - 12\frac{12\cdot 9}{16^2-12^2} - 7\frac{16\cdot 7-12^2}{16^2-12^2}) = 1.29.$

Estimation of MA process

Consider the MA(1) process

$$x_t - \mu = \epsilon_t + b_1 \epsilon_{t-1}.$$

The covariance function of this process is

$$\rho(k) = \begin{cases} 1 & k = 0, \\ \frac{b_1}{1+b_1^2} & k = \pm 1, \\ 0 & |k| > 1. \end{cases}$$

Th: The estimator of b_1 is given by a solution of the equation

$$\frac{\hat{b}_1}{1+\hat{b}_1^2} = \hat{\rho}(1)$$

such that $|\hat{b}_1| < 1$. Also, $\hat{\sigma}^2 = \hat{R}(0)/(1+\hat{b}_1^2)$.

Example MA(1)

Task. Let $x_1, \ldots, x_n$ be a time series from an MA(1) process with mean μ. Suppose that $\bar{X}_n = 2.3$, $\hat{R}(0) = 1.2$ and $\hat{\rho}(1) = 1/4$. Estimate parameters of this process: μ, b_1, σ^2.

We have $\hat{\mu} = 2.3$. Rewriting $\frac{\hat{b}_1}{1+\hat{b}_1^2} = 1/4$, we get

$$4\hat{b}_1 = 1 + \hat{b}_1^2.$$

That is, we have the quadratic equation

$$\hat{b}_1^2 - 4\hat{b}_1 + 1 = 0,$$

whose roots are $2 + \sqrt{3} \approx 3.73$ and $2 - \sqrt{3} \approx 0.27$.
Thus, the estimator of b_1 is given by $\hat{b}_1 = 2 - \sqrt{3}$ and the estimator of σ^2 is given by $\hat{\sigma}^2 = 1.2/(1 + (2 - \sqrt{3})^2) = 1.12$.

Homework: Learning with AI

Try the following prompts.

AI as a Mathematical Tutor

Prompt: "Explain the derivation of the Yule-Walker equations for an AR(2) process in simple terms. Why is the matrix $\hat{C}$ symmetric?"

Coding and Debugging

Prompt: "Modify the R code from Lecture 11 to estimate an AR(4) process and plot the original time series vs. the estimated residuals."

Self-Assessment

Task: Ask an AI to generate a synthetic time series of 10 points for an AR(1) process with $\mu = 5, a_1 = 0.7$. Then, perform the hand-calculations (as on slide 10) and ask the AI to verify your arithmetic.